

BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY

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भारतीय मानक मसौदा
हाइड्रोजन प्रौद्योगिकी — हाइड्रोजन सिस्टम की सुरक्षा के लिए बुनियादी मीमांसा
(IS 16749 का पहला पुनरीक्षण)

Draft Indian Standard
Hydrogen technologies — Basic Considerations for the Safety of
Hydrogen Systems
(First Revision of IS 16749)

ICS 27.075

Industrial Gases Sectional Committee, CHD 06

Last date for Comments: *See the cover page*

NATIONAL FOREWORD

(Formal clauses to be added later)

This standard provides general guidance on the safety aspects of hydrogen systems and technologies, including the properties, hazards, and risk mitigation measures associated with hydrogen. The standard is intended to promote the safe development and deployment of hydrogen applications and serves as a common reference for hydrogen-related safety information.

‘Hydrogen’ in this standard means protium (the most common isotope of hydrogen) (^1H), not deuterium (^2H) or tritium (^3H).

This standard was originally published in 2018 by identical adoption of ISO/TR 15916 : 2015. The committee responsible for the formulation of IS 16749 has decided to revise the Indian standard by adopting to ISO/TS 15916 : 2026.

The text of ISO Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain conventions are, however, not identical to those used in Indian Standards. Attention is particularly drawn to the following:

- Wherever the words ‘International Standard’ appears referring to this standard, they should be read as ‘Indian Standard’.
- Comma (,) has been used as a decimal marker, while in Indian Standards, the current practice is to use a point (.) as the decimal marker.

In this adopted standard, reference appears to certain International Standards where the standard atmospheric conditions to be observed are stipulated which are not applicable to tropical/subtropical countries. The applicable

standard atmospheric conditions for Indian conditions are $27\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$ and (65 ± 5) percent relative humidity and shall be observed while using this standard.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test or analysis, shall be rounded off in accordance with IS 2: 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

FOR COMPLETE TEXT OF THE DOCUMENT, KINDLY REFER ISO/TS 15916:2026

Note: The technical content of the document has not been enclosed as these are identical with the corresponding ISO Standard. For obtaining the copy of the complete ISO Standard, please contact:

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